

Prompt prefix for one-shot

In the speaker diarization transcript below, some words are potentially misplaced. Please correct those words and move them to the right speaker. For example, given this input transcript,

<spk:1> How are you doing today? I <spk:2> am doing very well. How was everything at the <spk:1> party? Oh, the party? It was awesome. We had lots of fun. Good <spk:2> to hear!

The correct output transcript should be:

<spk:1> How are you doing today? <spk:2> I am doing very well. How was everything at the party? <spk:1> Oh, the party? It was awesome. We had lots of fun. <spk:2> Good to hear!

Now, please correct the transcript below.\n

4.5 EVALUATION RESULTS

In Table 1, we show the evaluation results of the USM + turn-to-diarize baseline together with the outputs post-processed by DiarizationLM. We report results for zero-shot, one-shot, and finetuning on the diarization task with three different flavors.

For zero-shot and one-shot experiments with PaLM 2-S, we observe significantly worse WDER and cpWER performance compared with the baseline system, indicating the PaLM 2-S foundation model does not offer speaker diarization capabilities without finetuning. Zero-shot experiment with PaLM 2-L model also shows bad performance, while one-shot experiment with PaLM 2-L model is much better, but still worse than the baseline system. Our results indicate that the PaLM 2-L model with one-shot is able to improve speaker diarization in relatively simple cases as shown in Section 2 and Appendix B. However, real world applications can be much more complicated with errors from both the ASR system and the speaker diarization system. In such cases, even with one-shot, LLM can still introduce even more errors to the results if not finetuned specifically on the speaker diarization task.

On both datasets, we observe big improvement of both WDER and cpWER with any of the three finetuning flavors. Interesting, the biggest improvement is observed with the hyp2ora flavor, while the smallest improvement is observed with the deg2ref flavor. Specifically for hyp2ora, we see a rel. 55.5% improvement of WDER after post-processing with DiarizationLM on the Fisher testing set. Even if we did not use any Callhome data during the LLM finetuning, we see a rel. 44.9% improvement of WDER on the Callhome testing set. The WER of the USM on the two testing sets are relatively high due to domain mismatch and suboptimal annotation quality of the ground truth. However, this also demonstrated that the DiarizationLM solution provides consistent quality gains even with out-of-domain ASR and speaker diarization models.

To further demonstrate this, in Table 2, we show the results of a similar setup, but we replace the USM-based ASR model directly by the ground truth ASR transcripts from the testing sets. For these experiments, we will have WER=0%, and the hyp2ora and deg2ref flavors will be equivalent. From the table, we can still see big improvements of WDER after post-processing the diarization results by the same DiarizationLM model (i.e. deg2ref flavor in Table 1).

4.6 CASE STUDIES

Based on the results from Table 1, we also present example cases from the Fisher and Callhome testing sets where we see big improvements of WDER in Table 3 and Table 4, respectively. From these examples, we are seeing multiple patterns of corrections:

- DiarizationLM make corrections where **different parts of sentence** are moved to the same speaker, e.g. “it’s more of” and “it’ll be warm” in fe_03_07146 from Table 3. This is consistent with our initial observations as demonstrated in Section 2.
- DiarizationLM can merge short speaker turns due to **disfluency**, such as “yeah yeah” and “i i hear i hear” in fe_03_11159 from Table 3. Diarization errors from disfluency usually attribute to low quality speaker embeddings extracted from very short speaker turn segments.

Table 1: Evaluation results of the USM + turn-to-diarize baseline system and the results post-processed by DiarizationLM. For DiarizationLM, we experiment with PaLM 2 foundation models with and without finetuning on the diarization task. We also experiment with Llama 2 and Llama 3 models finetuned on the diarization task (model details in Appendix A). WERs are the same for all systems due to TPST. All numbers are percentages.

System	Fisher testing set			Callhome testing set		
	WER	WDER	cpWER	WER	WDER	cpWER
USM + turn-to-diarize baseline	15.48	5.32	21.19	15.36	7.72	24.39
+ PaLM 2-S zero-shot	-	11.96	30.19	-	12.26	30.60
+ PaLM 2-S one-shot	-	16.58	38.03	-	14.50	34.32
+ PaLM 2-L zero-shot	-	11.36	31.78	-	13.29	34.30
+ PaLM 2-L one-shot	-	5.94	22.21	-	7.95	24.67
+ PaLM 2-S finetuned (hyp2ora)	-	2.37	16.93	-	4.25	20.22
+ PaLM 2-S finetuned (deg2ref)	-	3.94	18.55	-	5.33	21.47
+ PaLM 2-S finetuned (mixed)	-	2.41	16.94	-	4.76	20.84
+ Llama 2 13B finetuned (mixed) v1	-	3.65	18.92	-	8.02	25.11
+ Llama 3 8B finetuned (mixed) v1	-	4.40	19.76	-	12.27	30.80
+ Llama 3 8B finetuned (mixed) v2	-	3.28	18.37	-	6.66	23.57

Table 2: Evaluation results of the turn-to-diarize baseline system with reference ASR transcript (assuming WER=0%) and the results post-processed by DiarizationLM. For DiarizationLM, we experiment with PaLM 2 foundation models with and without finetuning on the diarization task. All numbers are percentages.

System	Fisher testing set		Callhome testing set	
	WDER	cpWER	WDER	cpWER
Reference + turn-to-diarize baseline	2.81	5.19	3.74	6.82
+ PaLM 2-S zero-shot	7.50	12.70	7.29	12.79
+ PaLM 2-S one-shot	10.92	19.16	12.79	21.65
+ PaLM 2-L zero-shot	8.69	16.85	11.67	22.87
+ PaLM 2-L one-shot	3.23	5.99	3.76	6.95
+ PaLM 2-S finetuned	1.18	2.21	1.49	2.66

- DiarizationLM can also detect speaker turns due to **interruptions**, such as “oh all right” in fe_03_11210 from Table 3, and “oh my” in en_6408 from Table 4.

We also look into why zero-shot and one-shot experiments in Table 1 produced worse results than the baseline system. We found that without finetuning on the speaker diarization tasks, zero-shot and one-shot outputs from the LLM often delete big chunks of hypothesis text from the prompt. Finetuning the LLM is critical to avoid such undesired deletions. A few zero-shot examples with the PaLM 2-S model from the Fisher testing set were shown in Table 5.

5 DISCUSSION AND FUTURE WORK

The experiments in Section 4 have shown very promising results where LLMs can significantly reduce speaker diarization errors. However, we also admit the limitations of these experiments. First of all, the training and testing data from the experiments are all based on the telephone speech domain, all with exactly 2 speakers. An important future work would be to include more diverse datasets to finetune the LLM, and evaluate its performance across different domains with unknown number of speakers.

In Appendix B, we have demonstrated other diarization capabilities of LLMs. However, due to lack of relevant datasets, we haven’t been able to thoroughly evaluate these capabilities. One interesting future work would be to collect datasets of these tasks and evaluate how LLM performs.

Table 3: Example cases from the Fisher testing set where we see big absolute WDER reduction (Δ WDER) with DiarizationLM (deg2ref flavor).

Utterance	Before DiarizationLM	After DiarizationLM
fe_03_07146 (Δ WDER =8.80%)	... <spk:3> it's it's <spk:1> more of summer always like you know we never experience a bit cold over here <spk:4> usually it'll <spk:1> be warm or like very hot in summer yeah and <spk:3> extremely hot yeah with high humid- ity my humidity is pretty <spk:1> much high because i stay close to the sea coast over here <spk:3> yeah <spk:1> so <spk:3> that makes you live houston is it like houston where you live yeah i i i live <spk:1> in houston ...	... <spk:1> it's it's more of summer always like you know we never experience a bit cold over here usually it'll be warm or like very hot in summer <spk:2> yeah and extremely hot yeah with high humidity my <spk:1> humidity is pretty much high be- cause i stay close to the sea coast over here <spk:2> yeah so that makes you live houston is it like houston where you live <spk:1> yeah i i i live in houston ...
fe_03_06816 (Δ WDER =6.61%)	... <spk:3> uhuh <spk:2> did you see the the woman golfer that was on this the one <spk:1> monica yeah yeah <spk:2> what's her name monica stone yeah mhm she she <spk:1> blew out she fell out of that tourna- ment but i didn't think she'd do it she she's girls can't compete against guys ...	... <spk:2> uhuh did you see the the woman golfer that was on this the one <spk:1> monica yeah yeah <spk:2> what's her name monica stone <spk:1> yeah <spk:2> mhm <spk:1> she she blew out she fell out of that tournament but i didn't think she'd do it she she's girls can't compete against guys ...
fe_03_11210 (Δ WDER =6.35%)	... <spk:1> the vikings mine's the eagles i'm from new jersey oh all right i have my jersey on now i watch the game tonight yeah well i i may i may just watch <spk:2> part of it tonight too then but uh it's a case as i say if i had to pay for it i probably wouldn't watch it <spk:1> i wouldn't either uhuh <spk:2> unless <spk:1> it was an eagles game ...	... <spk:1> the vikings mine's the eagles i'm from new jersey <spk:2> oh all right <spk:1> i have my jersey on now i watch the game tonight yeah <spk:2> well i i may i may just watch part of it tonight too then but uh it's a case as i say if i had to pay for it i probably wouldn't watch it <spk:1> i wouldn't either <spk:2> uhuh <spk:1> unless it was an eagles game ...
fe_03_11159 (Δ WDER =4.05%)	... <spk:2> yeah <spk:1> anniversary that's horrible <spk:2> yeah <spk:1> yeah it's not good <spk:2> i <spk:1> i hear i hear you there that's not a good thing you <spk:2> know i mean of course you know that's a day that will go down instantly no- body will ever remember it ...	... <spk:1> yeah anniversary that's horrible yeah yeah it's not good i i hear i hear you there that's not a good thing <spk:2> you know i mean of course you know that's a day that will go down instantly no- body will ever remember it ...